



Global Innovation Hackathon 2026 –
Build for a Better Future

THE PATRON INDEX

WHERE BUYERS BECOME INSIGHTS

TEAM NAME -

CodeGoblins

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DOMAIN- BUSINESS & CUSTOMER ANALYTICS

PROBLEM STATEMENT

Businesses collect large amounts of customer purchase data, but often lack an effective way to understand and classify customers based on their buying behaviour.

Without proper customer segmentation, businesses may struggle to identify loyal, high-value, inactive, or at-risk customers, which can lead to ineffective marketing, poor customer retention, and missed revenue opportunities.

There is a need for a secure and intelligent platform that can transform historical purchase data into meaningful customer classifications and actionable business insights.



PROPOSED SOLUTION



The Patron Index is a secure customer intelligence platform that analyzes historical purchase data and automatically classifies customers into meaningful segments such as:

- High Value Customers
- Loyal Customers
- Regular Customers
- At-Risk Customers
- Inactive Customers



The platform uses factors such as purchase frequency, total spending, average order value, and recent activity to understand customer behaviour.

It provides business owners with an interactive dashboard containing customer classifications, analytics, trends, and recommendations, helping them make better decisions related to marketing, retention, and customer engagement.

Uploaded business data is also validated through a security layer before being processed.

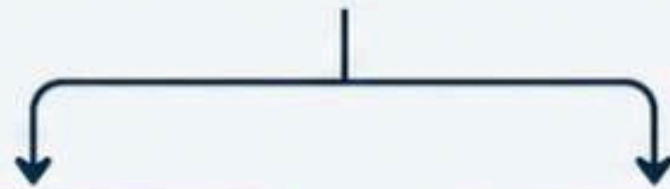


HOW THE PATRON INDEX WORKS



DATA INPUT

Business owner uploads customer purchase data through CSV or Excel files. The system collects transaction records for further processing.



UPLOAD FILE

CSV/Excel purchase data is added to the platform.

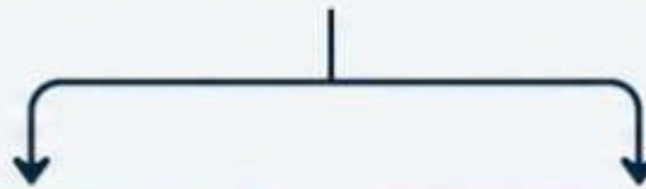
SECURITY CHECK

File type, structure, and suspicious content are validated before processing.



ANALYSIS

The platform studies buying behaviour using recency, frequency, spending, and average order value to understand each customer.



RFM METRICS

Recency, Frequency, and Monetary value are calculated.

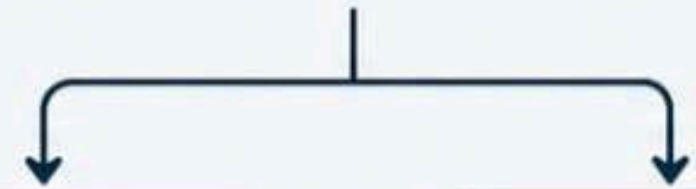
CLASSIFICATION

Customers are grouped as High-Value, Loyal, Regular, At-Risk, or Inactive.



OUTPUT

Customers are classified into useful groups and shown on a dashboard with insights and recommended actions.



DASHBOARD

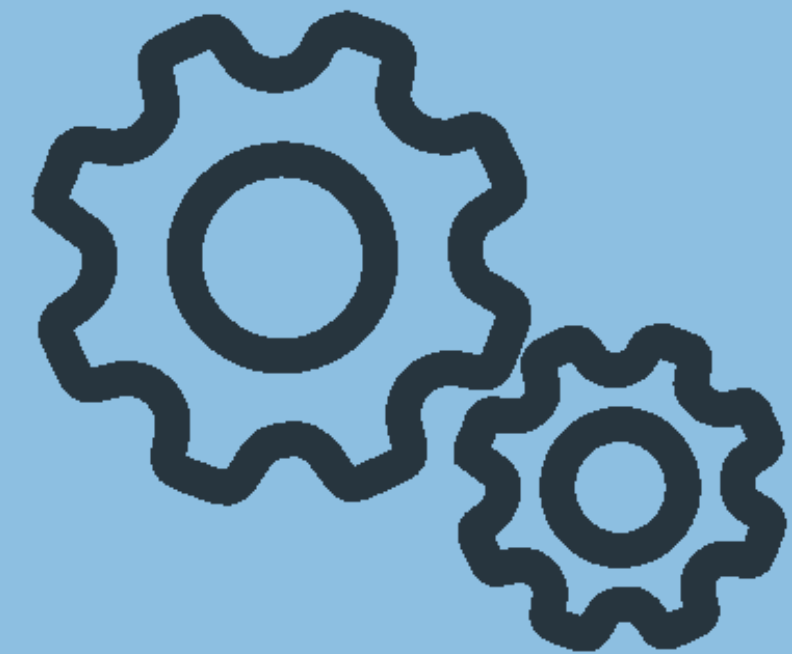
Segments, charts, and customer profiles are displayed in a simple view.

BUSINESS ACTION

Owners can plan retention, rewards, and marketing strategies based on the insights.

TECH STACK:

- 01** Frontend: React.js, Tailwind CSS, Chart.js
- 02** Backend: Node.js, Express.js
- 03** Database: MongoDB / My SQL
- 04** Machine Learning: Python, Scikit-learn, Pandas
- 05** Security: Malware Detection + File Validation
- 06** ML API : Fast API
- 07** Version Control : Git, Github



INNOVATION & TARGET USERS



WHAT MAKES IT DIFFERENT

The Patron Index transforms raw purchase data into clear customer intelligence that business owners can actually use.

01



Smart Classification

Automatically groups customers as High-Value, Loyal, Regular, At-Risk, and Inactive.

02



Actionable Insights

Turns customer segments into useful retention, reward, and targeting decisions.

03



Secure Processing

Uploaded CSV or Excel files are validated before analysis for safer data handling.

04



Simple for Owners

A clean dashboard and easy file upload make it usable for non-technical users.



TARGET USERS

Built for businesses that want better customer understanding without complex analytics tools.



Small Businesses

Local stores, cafés, boutiques, salons, and homegrown brands.



Medium Businesses

Growing retailers, restaurant chains, service brands, and regional businesses.



E-Commerce & D2C

Online sellers aiming for better segmentation, retention, and repeat purchases.



Marketing & Sales Teams

Teams that need clear customer groups for offers, campaigns, and follow-ups.

FEASIBILITY & SCALABILITY

How we'll build it now and expand it later.



FEASIBILITY

The Patron Index can be built as a practical MVP using a modular architecture and commonly used tools.



MVP FIRST

Start with CSV upload, customer analysis, basic classification, and a clean dashboard.



PRACTICAL STACK

Use React, Node.js, MongoDB, and a Python ML service for fast development.



MODULAR BUILD

Frontend, backend, analytics, and security modules can be developed separately and integrated smoothly.



DEMO READY

A working prototype can be shown with sample purchase data, segments, and insights.



SCALABILITY

Once the MVP works, the platform can grow into a larger business intelligence system.



MORE DATA

Handle larger transaction volumes and support recurring uploads from businesses.



MORE USERS

Support multiple businesses, team logins, and role-based access.



SMARTER INSIGHTS

Add churn prediction, loyalty scoring, and campaign recommendations.



PLATFORM EXPANSION

Integrate with POS systems, e-commerce tools, and future reporting modules.

EXPECTED IMPACT

How The Patron Index helps businesses make smarter customer decisions



RETENTION

Identify at-risk customers early and improve repeat purchases through timely action.



TARGETED MARKETING

Create better offers, rewards, and campaigns for the right customer groups.



CUSTOMER UNDERSTANDING

Understand which customers are loyal, high-value, regular, or inactive.

BUSINESS OUTCOMES



Better customer loyalty



Smarter campaign planning



Improved decision-making

PROJECT REFERENCES

SECURITY MODULE

H. S. Anderson and P. Roth, "EMBER: An Open Dataset for Training Static PE Malware Machine Learning Models," **arXiv:1804.04637**, 2018.
<https://arxiv.org/abs/1804.04637>

EMBER dataset and feature-extraction code (Elastic).
<https://github.com/elastic/ember>

E. Raff et al., "Malware Detection by Eating a Whole EXE" (MalConv, CNN on raw bytes), 2017.
<https://arxiv.org/abs/1710.09435>

C. Guo et al., "On Calibration of Modern Neural Networks," **ICML**, 2017.
<https://arxiv.org/abs/1706.04599>

PyTorch.
<https://pytorch.org/docs>

CUSTOMER INTELLIGENCE & ML

scikit-learn: K-Means, Gaussian Mixture Models, clustering.
<https://scikit-learn.org/stable/modules/clustering.html>

G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," **NeurIPS**, 2017.
<https://lightgbm.readthedocs.io>

T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," **KDD**, 2016.
<https://arxiv.org/abs/1603.02754>

S. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions" (SHAP), **NeurIPS**, 2017.
<https://arxiv.org/abs/1705.07874>

SHAP documentation.
<https://shap.readthedocs.io>

APP STACK & DEPLOYMENT

FastAPI.
<https://fastapi.tiangolo.com>

React.
<https://react.dev>

Node.js / Express.js.
<https://expressjs.com>

MongoDB.
<https://www.mongodb.com/docs>

Docker.
<https://docs.docker.com>



THANK YOU

~ Team CodeGoblins